

Exercise 1. Compute last two digits of 7^{2025} .

Exercise 2. Compute $3^{123456789} \pmod{164}$.

Exercise 3. Compute multiplicative inverse of 3 $\pmod{97}$, that is an integer n such that $3n \equiv -1 \pmod{97}$. Also, compute $3^{95} \pmod{97}$. Why are they related?

Exercise 4. Assume that a is primitive modulo a prime p (that is order of a is $p-1$). Show that $a^{pq} \not\equiv a \pmod{pq}$ for any prime $q < p$.

Exercise 5. Show that 2 is primitive root modulo 83 by explicitly showing $2^{41} \not\equiv 1$ and $2^2 \not\equiv 1 \pmod{83}$. Now what are all the primitive roots $\pmod{83}$ written as powers of 2?

Exercise 6. Let $F, G : \mathbb{N} \rightarrow \mathbb{C}$ be arithmetic functions. Recall that we defined $F * G := \sum_{d|n} F(d)G(n/d)$. Show that:

$$\mu * 1 = \delta_1$$

where 1 is the constant function given by $1(n) = 1$ for all n , δ_1 is the dirichlet delta function given by

$$\delta_1(n) = \begin{cases} 1 & \text{if } n = 1 \\ 0 & \text{otherwise} \end{cases} \text{ and } \mu \text{ is the function by}$$

$$\mu(n) = \begin{cases} 1 & n = 1 \\ (-1)^k & n = p_1 p_2 \cdots p_k \text{ for } p_i \neq p_j \text{ distinct primes.} \end{cases}$$

Hint: If $n = 1$, LHS is explicit. For $n = p_1^{a_1} \cdots p_r^{a_r}$, non-zero terms on the LHS involve $d = p_{i_1} p_{i_2} \cdots p_{i_k}$ for $i_s \neq i_t$.

Exercise 7 (Möbius Inversion Formula). Use the previous exercise to show that if $F(n) = \sum_{d|n} G(d)$ then $G(n) = \sum_{d|n} \mu(d)F(n/d)$.

*Hint: Show that $\delta_1 * G = G$ and use associativity of $*$.*

Exercise 8. Let $F, G : \mathbb{N} \rightarrow \mathbb{C}$ be arithmetic functions. Show that:

- (1) Recall that an arithmetic function is multiplicative if $F(mn) = F(m)F(n)$ whenever $\gcd(m, n) = 1$. Show that if F, G are multiplicative, so is $F * G$.
- (2) Recall that an arithmetic function is totally multiplicative if $F(mn) = F(m)F(n)$ for all $m, n \in \mathbb{N}$. Show by example that there are totally multiplicative F, G such that $F * G$ is not so.

Exercise 9 (Perfect Numbers).

- (1) Let $\sigma(n) := \sum_{d|n} d$, that is sum of divisors of n . Show that σ is multiplicative. Conclude that

$$\sigma(p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}) = \left(\frac{p_1^{a_1+1} - 1}{p_1 - 1} \right) \cdots \left(\frac{p_r^{a_r+1} - 1}{p_r - 1} \right).$$

- (2) A number is perfect if it is equal to sum of its divisors. Show that if n is even, then n is perfect iff $n = 2^{p-1}(2^p - 1)$ for some prime p such that $2^p - 1$ is a prime.

Hint: A number is perfect iff $\sigma(n) = 2n$. Now if $n = 2^k m$ for m odd and $k \geq 1$, $\sigma(n) = \sigma(2^k)\sigma(m) = (2^{k+1} - 1)\sigma(m) = 2^{k+1}m$. It follows that $2^{k+1} - 1$ divides m , and hence $2^{k+1} \frac{m}{2^{k+1} - 1} = \sigma(m) \geq m + \frac{m}{2^{k+1} - 1} = 2^{k+1} \frac{m}{2^{k+1} - 1}$. Conclude that $2^{k+1} - 1$ should be a prime, and then that p is a prime.